

END TERM EXAMINATION (AUTUMN SEMESTER)-2023
ALIAH UNIVERSITY

Course code: PHYUGMCC1101

Course name: Mechanics

Department of Appearing Students: Physics Department

Programme Name: B.Sc. (Physics Honours)

Year: 1st Year

Semester: I

Full Marks: 60

Time: 02:30 Hrs

Attempt questions as directed.
(Use separate Answerscripts for Group-A and Group-B answers.)

Group-A: [Answer any **three** of the following questions. 10 × 3 = 30]

1. (a) Calculate the gravitational potential and hence intensity at any point inside the hollow sphere of radius R.
(b) If the value of the gravitational potential at the center of the earth becomes zero, then what will be the value of the potential at the surface of the earth and infinity?
(c) Plot the variation of gravitational intensity and potential due to a solid sphere with distance from its center. [6+2+2=10]
2. (a) Prove that $Y = 3K(1 + 2\sigma)$, where symbols bear their usual meaning.
(b) What is flexural rigidity? Find the expression for depression for a loaded heavy cantilever.
(c) A steel rod of length 50 cm, width 2 cm, and thickness 1 cm is bent into the form of an arc of a radius of curvature 2.0 m. Calculate the internal bending moment of the rod. Young's modulus of the material of the rod $= 2 \times 10^{11} \text{ N/m}^2$. [4+4+2=10]
3. (a) Deduce Bernoulli's theorem for the flow of liquid through a narrow tube using Euler's equation.
(b) Show that the time required to empty a tank $t = \frac{A}{\alpha} \sqrt{\frac{2H}{g}}$, where A and α are the area of the tank and the small hole, and H is height of the liquid column.
(c) The height of liquid inside a cylindrical vessel is H . Show that the range where the liquid strikes the ground will be maximum when the hole is at a height $H/2$ from the bottom of the vessel. [4+3+3=10]
4. (a) What is ideal fluid? Deduce Poiseuille's formula for the flow of liquid through a narrow horizontal tube. .
(b) A uniform square plate of side 'a' and mass 'm' is hung up by one corner and swings in the vertical plane. Find the time period of it. [(1+6)+3=10]

Group-B: [Answer any **three** of the following questions. 10 × 3 = 30]

5. (a) Discuss the principle of rocket motion. Hence, derive an expression for the acceleration of a rocket neglecting the influence of gravity.
(b) A rocket starts vertically upward with a speed v_0 . Show that its speed at a height h given by $v^2 = v_0^2 - \frac{2gh}{1+h/R}$, where R is the radius of earth and g is the acceleration due to gravity at the surface of the earth. [(2+4)+4=10]
6. (a) Obtain expressions for the radial and tangential (transverse) components of i) velocity and ii) acceleration of a particle using plane polar coordinates.

- (b) What do you mean by inverse square law of forces? Check whether such forces are conservative or not. [(2+4)+(2+2)=10]
7. (a) What are central and non-central forces? Give an example of each.
- (b) Show that the differential equation of motion of a particle of mass m moving in a central force field can be written as $\frac{d^2 u}{d\theta^2} + u = -\frac{m}{L^2 u^2} F\left(\frac{1}{u}\right)$
- (c) Show that the mechanical energy of an object moving under a central force is conserved. [(2+1)+4+3=10]
8. (a) What are the conservative and non-conservative forces? Give an example of each.
- (b) State and explain the Work-Energy theorem. A particle of mass m is projected to a height n times the radius of the earth, calculate the work done.
- (c) A neutron (mass 1.0 u) traveling at 2.7×10^7 m/s undergoes a head-on elastic collision with a Uranium nucleus (mass 235 u) initially at rest. Find the velocities of the particles after collision. What amount of energy will be transferred from the neutron to the uranium nucleus in this collision? Given: 1 u is the atomic mass unit equalling 1.66×10^{-27} kg. [2+(2+2)+(2+2)=10]
9. (a) State Kepler's laws of planetary motion.
- (b) Derive Newton's law of gravitation from Kepler's laws.
- (c) The minimum and maximum distance of a planet from the sun are 7×10^{10} m and 1.4×10^{12} m respectively. If the speed of the planet at the nearest position is 6×10^4 m/s, then what will be its speed at the farthest position? [3+4+3=10]